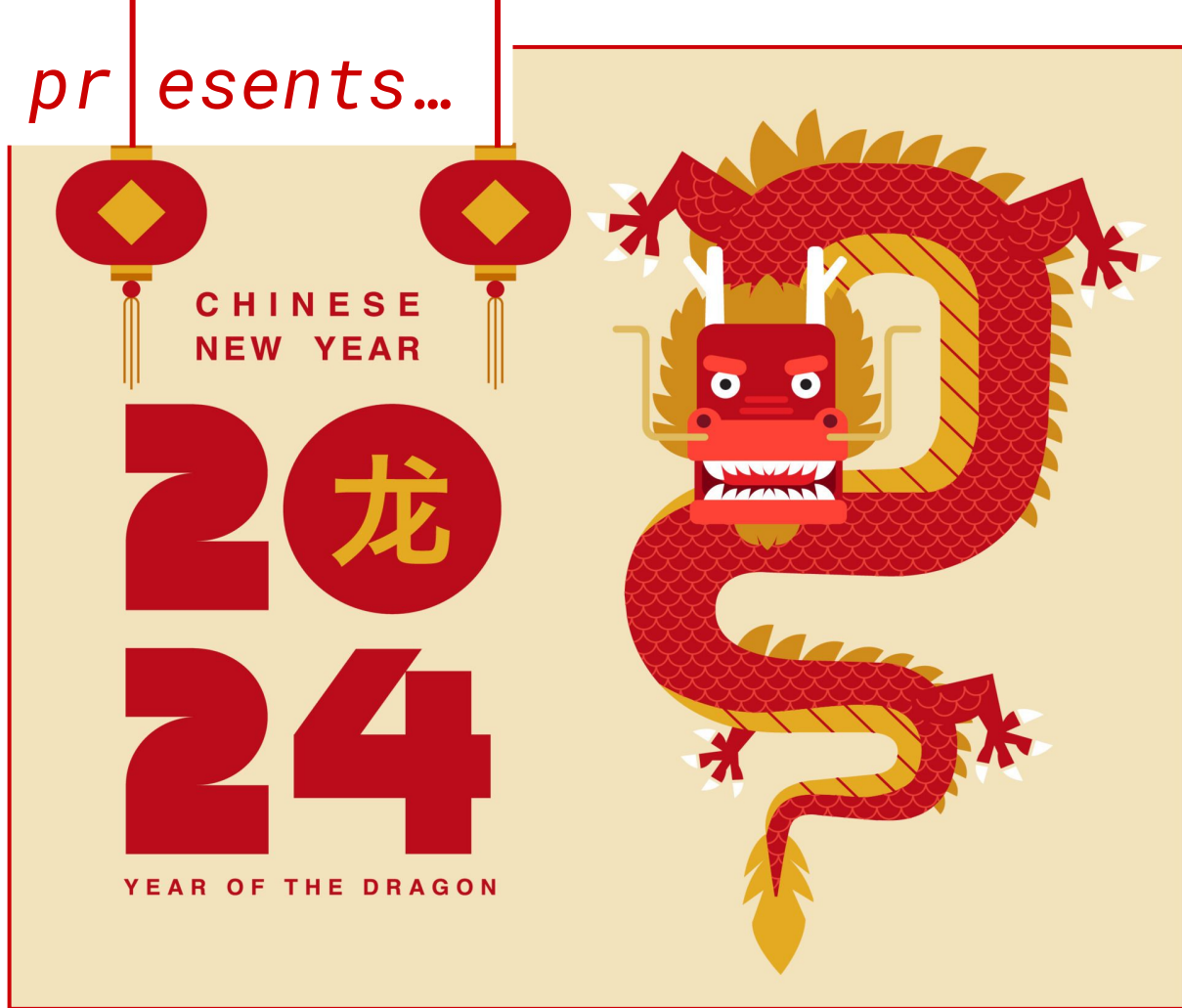


Chemistry Club presents...

With: flame tests.

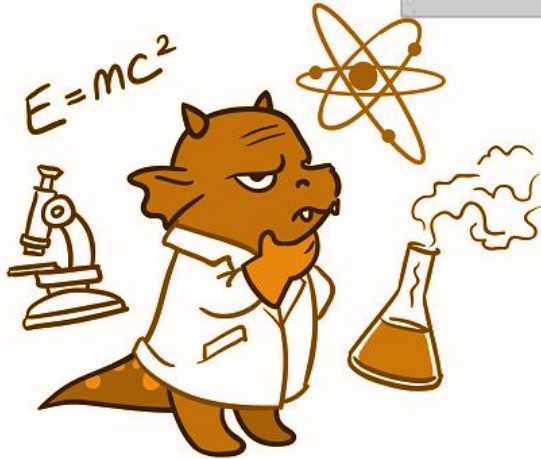
Hosted by: This dragon who is
also a super skilled scientist
- Dr. Dragon.

"Why are scientists always thinking about
 $E = mc^2$?" - My Brother 2024



What are flame tests?

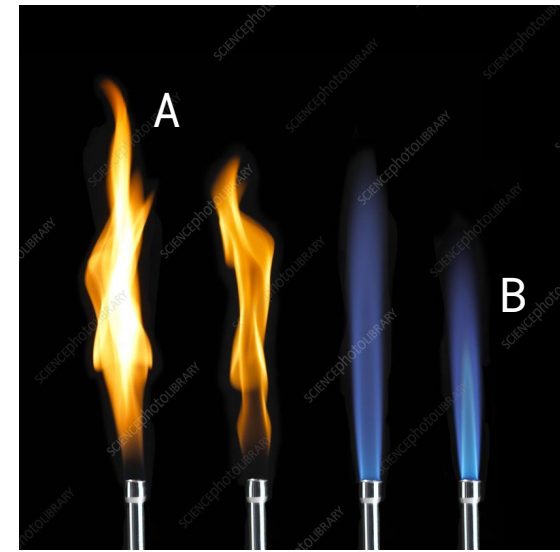
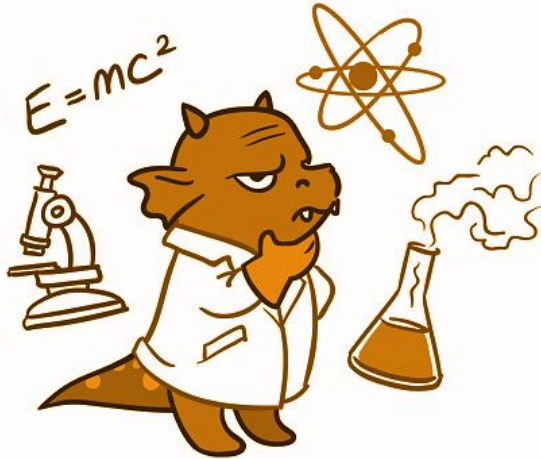
Flame tests are used by analytical chemists (me) to identify the presence of certain **ions** in another dragon's breath.



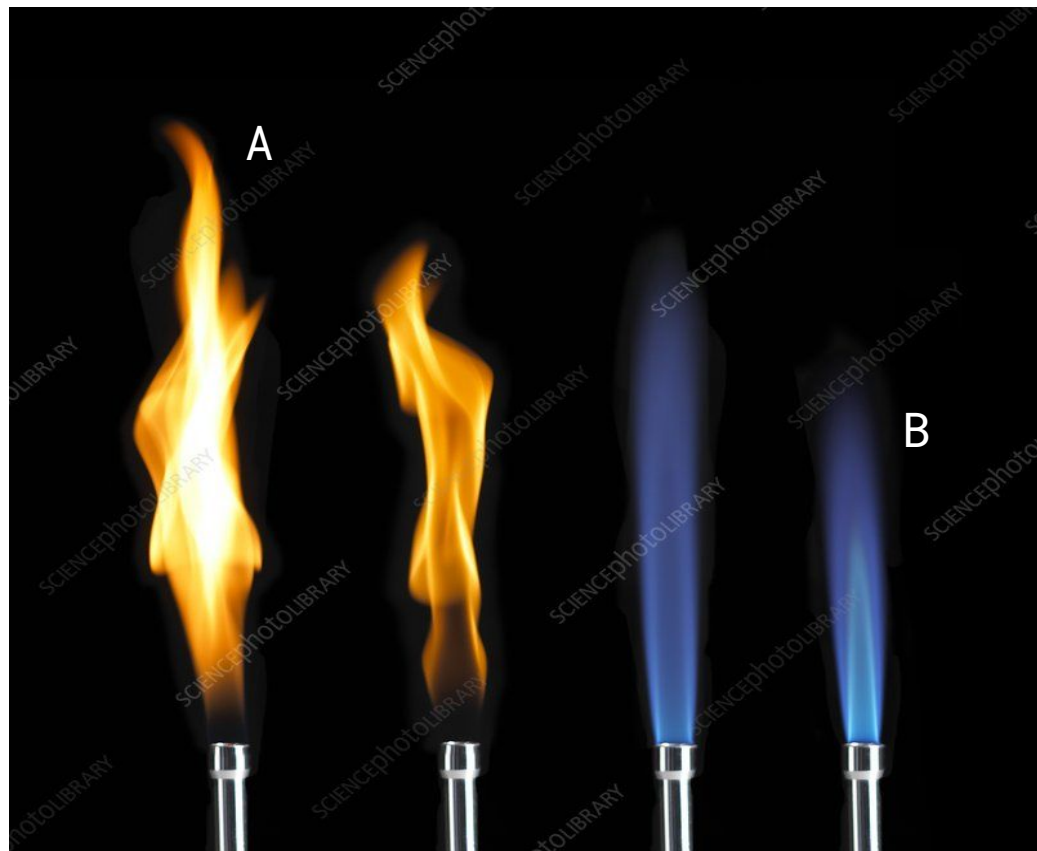
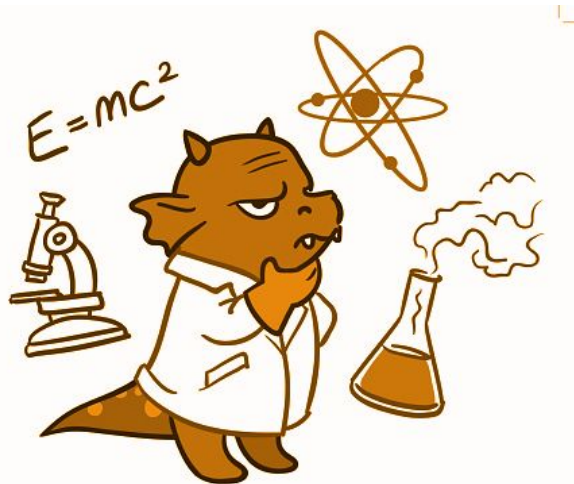
Ion - Atoms or groups of atoms with charge.

Safety

Fire very dangerous. I am a dragon I would know. Be safe.

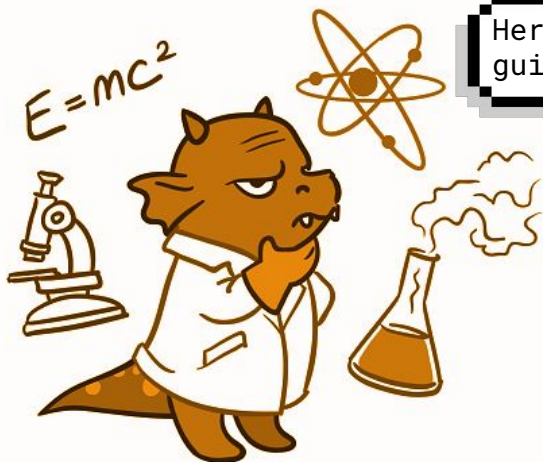


- A. **Safety flame** (whole closed)
- B. **Roaring flame** (whole open) - use only when testing the sample.



Flame Tests

Element	Ion	Flame test colour
Lithium	Li ⁺	Crimson
Sodium	Na ⁺	Yellow
Potassium	K ⁺	Lilac
Calcium	Ca ²⁺	Orange-red
Copper	Cu ²⁺	Green

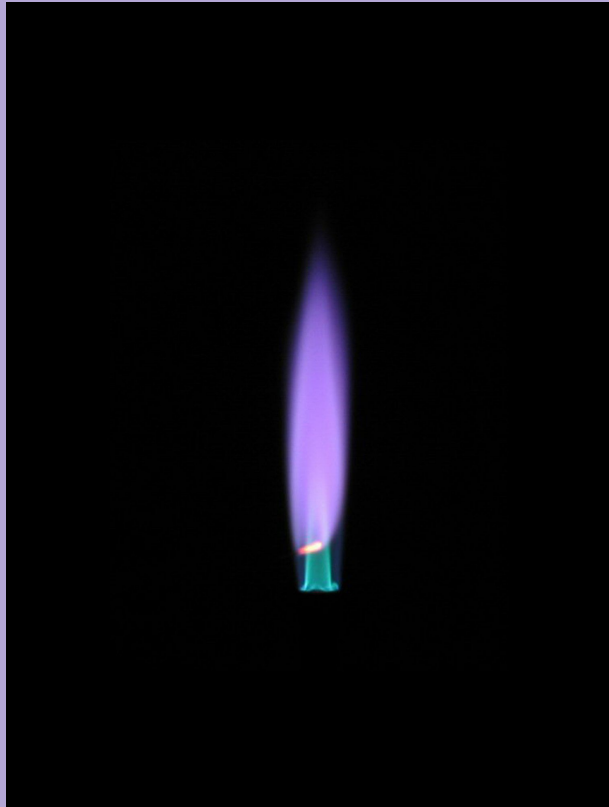


Here is my very detailed guide to flame tests...

Method:

1. Set up equipment.
2. Dip splint in water.
3. Dip splint in sample.
4. Add sample to roaring bunsen burner flame.
5. Observe colour change.
6. Identify the ion.

Potassium, K^+



Copper, Cu^{2+}



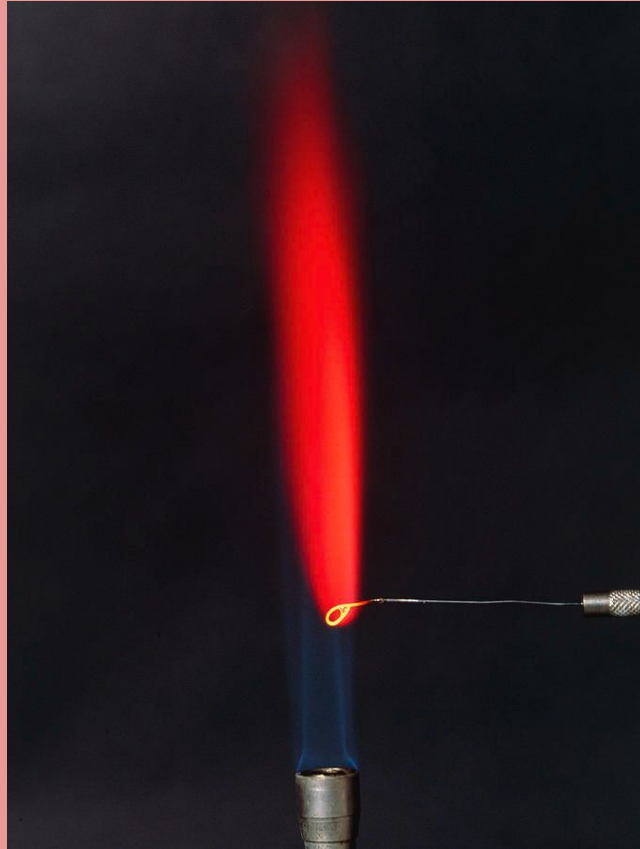
Calcium, Ca^{2+}



Sodium, Na^+



Lithium, Li^+ (my fav!)

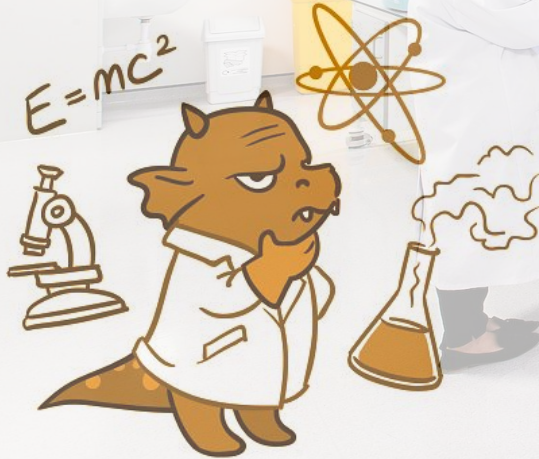


Welcome to Dr. Dragon's lab...

When you heat the sample electrons absorb energy.

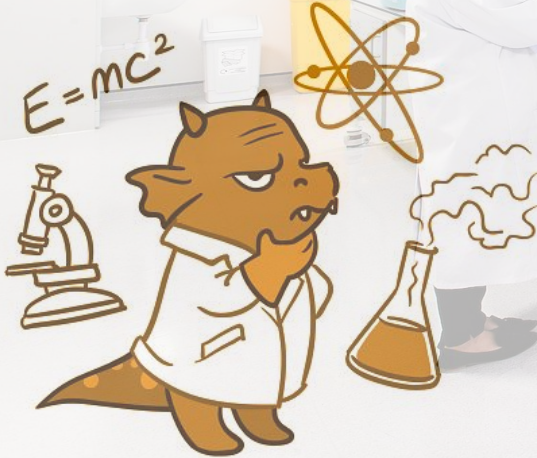
They become excited and jump further from the nucleus.

They then "calm down" and move back. This releases visible light.



Welcome to Dr. Dragon's lab...

Electrons in different atoms, release different wavelengths of visible light so they appear different **colours**.



Welcome to Dr. Dragon's lab...

